Bayes factors

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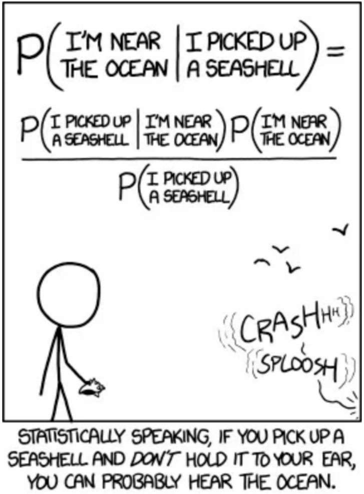
From Bayes theorem to Bayes factors



https://xkcd.com/1236/

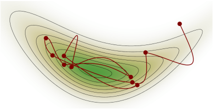
Bayes theorem

Follows directly from the axioms of (classical) probability theory

- P(D & H) = P(D)P(H|D)
- P(D & H) = P(H)P(D|H)
- P(D)P(H|D) = P(H)P(D|H)
- P(H|D) = P(H)P(D|H) / P(D).

(Joyce, 2003).

Bayesian conditionalization

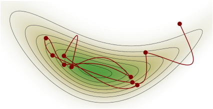


Generalizes Bayes theorem from static to dynamic setting, allowing inference from prior to posterior state of affairs.

- Bayes theorem: $P_1(H|D) = \frac{P_1(H)P_1(D|H)}{P_1(D)}$.
- Bayesian conditionalization: $P_2(H) = P_1(H|D)$.
  - P_1 = probability at time 1; P_2 = probability at time 2.
  - Presupposes the data D is learned with certainty (but see also Jeffrey conditionalization).
  - Presupposes invariance in the conditional probabilities between time points: $P_1(H|D) = P_2(H|D)$.
  - When assumptions met and priors internally consistent, then repeated application of Bayesian conditionalization tends to lead to convergence with the ground truth (where there is one).

(Hájek, 2023; Joyce, 2003).

Bayesian conditionalization



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(Hájek, 2023; Joyce, 2003).

Comparing hypotheses

By Bayes theorem:

$$P(H_1|data) = \frac{P(data|H_1)P(H_1)}{P(data)}$$

$$P(H_0|data) = \frac{P(data|H_0)P(H_0)}{P(data)}$$

By Bayesian conditionalisation:

- $P_1(H_1|data) = P_2(H_1)$: posterior probability of H_1 after observing the data
- $P_1(H_0|data) = P_2(H_0)$: posterior probability of H_0 after observing the data

Relative evidence

Evidence for H_1 relative to H_0 given the data:

$$\frac{P(H_1|data)}{P(H_0|data)} = \frac{P(data|H_1)}{P(data|H_0)} * \frac{P(H_1)}{P(H_0)}$$

Posterior odds = Bayes Faktor * Prior odds

- Posterior odds: Evidence for the hypothesis after seeing the data
- Bayes factor (BF): Amount of evidence that the data provide for one hypothesis relative to another hypothesis
- Prior odds: Evidence for the hypothesis before seeing the data

Relative evidence

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Posterior odds $=$ Bayes Faktor $*$ Prior odds

- Posterior odds: Evidence for the hypothesis after seeing the data
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Bayes factors (*BFs*)

$$BF_{12} = \frac{P(data|H_1)}{P(data|H_0)}$$

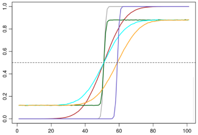
- BF_{10} = Probability of the data given H_1 relative to the probability of the data given H_0.
- Corresponds to the relative evidence for H_1 relative to H_0. Values above 1: Evidence for H_1. Values below 1: Evidence for H_0. Values near 1: The data is not that informative, no clear evidence for or against H_1.
- Instead of Hypotheses H we could refer to any two models M_a and M_b that we wish to compare.

BFs: Ratios of marginal likelihoods

$$BF_{ab} = \frac{P(data|M_a)}{P(data|M_b)}$$

- BF_{ab} is the ratio of the marginal likelihood of model M_a to the marginal likelihood of Model M_b.
- This corresponds to the relative evidence the data provide for M_a relative to M_b.
- Marginal likelihoods do not depend on a specific parameter value: all possible prior parameter values are taken into account simultaneously.

Strength of evidence



- Cutoff values have been proposed to categorise *BF*s according to the strength of evidence they provide.
- These are just suggestions for when no more specific evidence strength criteria (informed by research questions and goals) are available.
- Some Bayesians reject cutoff values altogether, preferring to report the quantitative evidence provided by the *BF* directly.
- Alternative: combine the continuous evidence of *BF*s with discrete decision thresholds to maximise expected utility of acting on a particular hypothesis.

(Nicenboim, Schad, & Vasishth, 2025;Veenman et al., 2023).

Jeffrey's evidence thresholds

Table 7.1 Evidence categories for the Bayes factor BF_{12} (Jeffreys, 1961).

Bayes factor BF_{12}			Interpretation
	$>$	100	Extreme evidence for $\mathcal{M}_1$
30	–	100	Very strong evidence for $\mathcal{M}_1$
10	–	30	Strong evidence for $\mathcal{M}_1$
3	–	10	Moderate evidence for $\mathcal{M}_1$
1	–	3	Anecdotal evidence for $\mathcal{M}_1$
	1		No evidence
1/3	–	1	Anecdotal evidence for $\mathcal{M}_2$
1/10	–	1/3	Moderate evidence for $\mathcal{M}_2$
1/30	–	1/10	Strong evidence for $\mathcal{M}_2$
1/100	–	1/30	Very strong evidence for $\mathcal{M}_2$
	$<$	1/100	Extreme evidence for $\mathcal{M}_2$

(Lee & Wagenmakers, 2013).

Some properties of Bayes factors

$$BF_{ab} = \frac{P(data|M_a)}{P(data|M_b)}$$

$$BF_{ba} = 1/BF_{ab}$$

$$BF_{ac} = BF_{ab} * BF_{bc}$$

- *BFs* offer quantitative evidence for one model relative to another.
- The evidence is not symmetrical ($BF_{ab} \neq BF_{ba}$); but knowing the evidence for *a* relative to *b* is enough to compute the evidence for *b* relative to *a*.
- *BFs* are transitive.
- *BFs* are odds ratios (OR), not probabilities.

(Rouder & Morey, 2018; Veenman et al., 2023).

Bayes factors and parsimony

Example

Holmes and Watson play a simple game of darts, in which, with each dart, the player tries to score as many points as possible. The maximum score per dart is 60, and the minimum score is 0 (when the dart lands outside the board). After 5 darts, Holmes scored 38, 10, 0, 0, 0and Watson scored 20,20,20,18,16. How do you determine who is the better player? Consider another game of darts, but now one of the players gets to throw 50 times instead of 5. This scenario illustrates the importance of averaging instead of maximizing in order to penalize complexity.

(Lee & Wagenmakers, 2013, Ch. 7).

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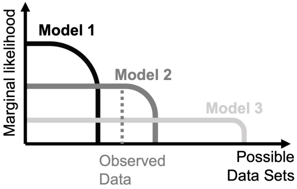
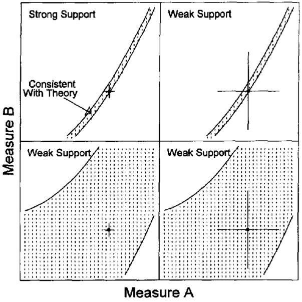
BFs account for differences in model flexibility



- A model has high marginal likelihood if it makes high proportion of good predictions.
- Model predictions are normalized: the total probability that models assign to different data patterns is the same for all models.
- A model that predicts many different outcomes spreads its prior predictive density more thinly among them. This flexibility gives the model higher chances of predicting the actually observed outcome; but due to normalization it cannot predict it with high probability.

(Lee & Wagenmakers, 2013, Nicenboim, Schad, & Vasishth, 2025; Veenman et al., 2023).

Complexity and informativeness



- More complex, flexible models are less informative and more difficult to falsify → less useful as scientific theories.
- Models are more complex when they have (a) wider priors, (b) more parameters, (c) less constrained functions.
- Model comparison with *BFs* penalizes unnecessarily complex models (Occam's razor).

(Roberts & Pashler, 2000).

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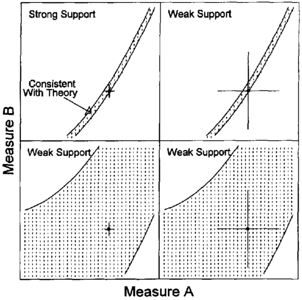

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Methods for computing Bayes factors

Two main estimation algorithms

- Savage-Dickey density ratio
- Bridge sampling

(Veenman et al., 2023).

Savage-Dickey density ratio (SD density-ratio)

- Estimates the mean effect of a parameter by comparing nested models with vs. without the parameter.
  - → E.g. M_a estimates the distribution of a parameter, while in M_b the parameter is set to zero or to another constant.
- Die models compared include the same effects of interindividual differences, making them similar to type III sum os squares in ANOVA.
- Relatively efficient, computable analytically in special cases.
- In `brms`: e.g. via `hypothesis()` function of the same package.

(Veenman et al., 2023; Wagenmakers et al., 2010).

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SD density-ratio: limitations

- Estimation quality depends on the density of the parameter distribution at the point of comparison (e.g. zero). If the distribution has low density at that point (e.g. when the effect of the parameter is large) then its estimation becomes unreliable.
  - $\rightarrow$ Can be partially compensated for by drawing more samples
- Limited to nested models that differ in a single parameter.

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Bridge sampling

- More general and versatile than SD density-ratio.
- Applicable to non-nested models and to models that differ in more than one parameter.
- Based on sampling the posterior distribution of a model to obtain the corresponding marginal likelihood of the data.
- The marginal likelihoods of the data given different models can then be directly compared.
- In brms: e.g. via `bayes_factor()` from `multibridge` package, or `bayesfactor_models()` from `bayestestR` package.

(Gronau et al., 2017; Veenman et al., 2023).

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Bridge sampling: Limitations

- Less efficient than SD density ratio.
- Like all sampling based methods it offers an estimate, not an exact value. *BF* estimation typically requires more samples than the estimation of posterior model parameters.
  - → Rule of thumb: 10 x as many iterations
  - → Stability analysis by repeatedly running the model, and for each model run repeatedly computing the *BF* (e.g. 5 x 5 times).

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Sensitivity analysis for *BFs*

Types of sensitivity analysis

- Assess to what extent the analysis results depend on prior assumptions and modelling decisions, e.g. about
  - prior parameter means
  - prior parameter variances
  - distribution family
  - number of samples, sampling algorithm

(Veenman et al., 2023).

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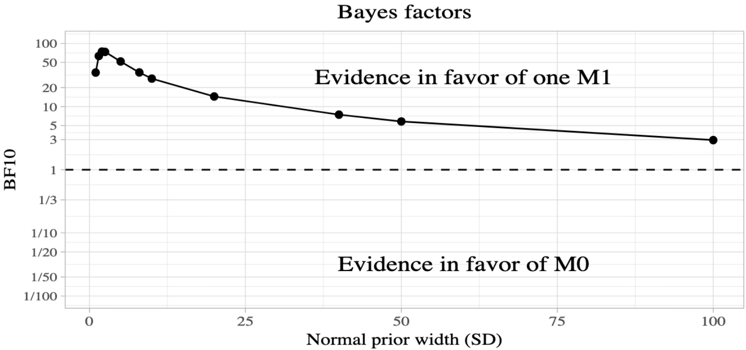
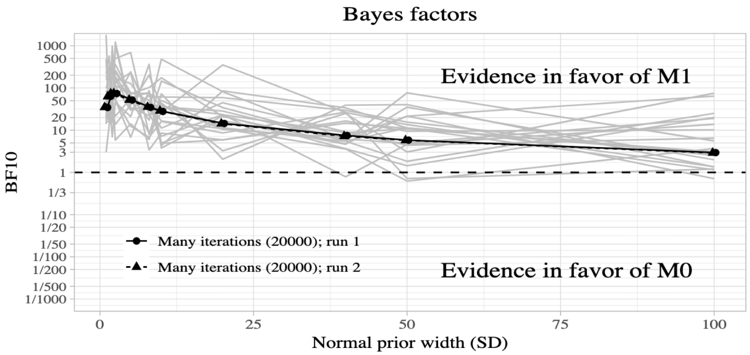
BF sensitivity to prior *SD*



- Robust evidence for effect, but effect size probably small
- Wider priors on *SD* reduce evidence for effect

(Nicenboim, Schad, & Vasishth, 2025). *Note.* SD priors: [1, 1.5, 2, 2.5, 5, 8, 10, 20, 40, 50, 100].

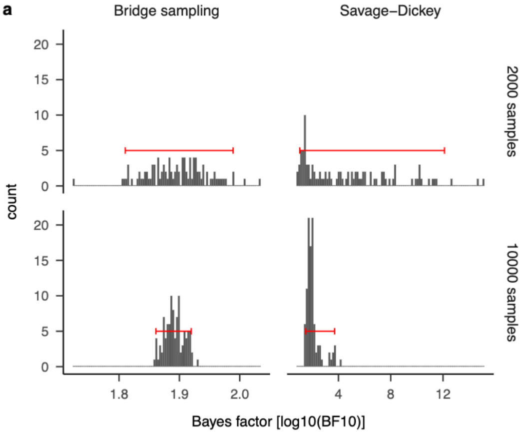
BF sensitivity to iteration number



- Stable *BF* estimates require more samples than stable parameter estimates.

(Nicenboim, Schad, & Vasishth, 2025). *Note.* Black lines: 2 runs with 20 000 iterations. Gray lines: 20 runs with default number of iterations (2000).

BF sensitivity to estimation method



(Schad et al., 2023). *Note.* Red horizontal error bars indicate 95 percent quantiles.

Example with brms

Penguin example

To what extent can be a penguin's body mass be predicted from the length of its bill and of its flippers?

```
> library(brms)
> m1 <- brm(body_mass ~ bill_len + flipper_len
            , data = penguins
            )
```

BF comparison with brms: Savage-Dickey (1)

```
prior_2 <- prior(normal(0, 5), class = b)

m1 <- brm(body_mass ~ bill_len + flipper_len
                      , data = penguins
                      , warmup = 1000
                      , iter = 6000
                      , chains = 4
                      , cores = 4
                      , prior = prior_2
                      , sample_prior=TRUE
                      , save_pars = save_pars(all = TRUE)
                     )

m0 <- brm(body_mass ~ flipper_len
          , data = penguins
          , warmup = 1000
          , iter = 6000
          , chains = 4
          , cores = 4
          , prior = prior_2
          , sample_prior=TRUE
          , save_pars = save_pars(all = TRUE)
           )
```

Note: To make sure the computations work, specify proper priors for the predictors, and include *sample_prior=TRUE* and save_pars = save_pars(all = TRUE) in the call to `brm()`.

(Speekenbrink, 2023).

BF comparison with `brms`: Savage-Dickey (2)

```
# BFs via Savage-Dickey
> h <- hypothesis(m1, "bill_len = 0", class = "b")

Hypothesis Tests for class b:
      Hypothesis Estimate Est.Error CI.Lower CI.Upper Evid.Ratio
1 (bill_len) = 0   8.2245    3.5614   1.1489  15.1791     0.1095
  Post.Prob Star
1    0.0987    *
---
'CI': 90%-CI for one-sided and 95%-CI for two-sided hypotheses.
Posterior probabilities of point hypotheses assume equal prior probabilities.

> bf_12 <- h$hypothesis$Evid.Ratio
> bf_12
[1] 0.1095364

> bf_21 <- 1 / bf_12
> bf_21
[1] 9.129385
```

The data was accounted for around 9 times better under the assumption that both bill length and flipper length predict body mass (m_1), compared to the assumption that only flipper length predicts body mass (m_0).

BF comparison with Bridge sampling

```
> prior_2 <- prior(normal(0, 5), class = b)
> m1 <- brm(body_mass ~ bill_len + flipper_len
                      , data = penguins
                      , warmup = 1000
                      , iter = 6000
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                      , cores = 4
                      , prior = prior_2
                      , sample_prior=TRUE
                      , save_pars = save_pars(all = TRUE)
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> m0 <- brm(body_mass ~ flipper_len
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              , warmup = 1000
              , iter = 6000
              , chains = 4
              , cores = 4
              , prior = prior_2
              , sample_prior=TRUE
              , save_pars = save_pars(all = TRUE)
          )
# Using bridge sampling
> bayes_factor(m0, m1)
Estimated Bayes factor in favor of m1 over m0: 20.57544
```
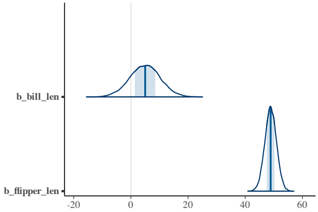
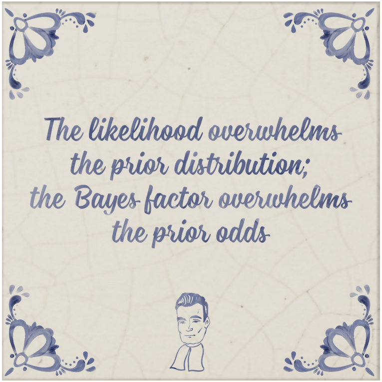

The data was around 20 times more likely under the assumption that both bill length and flipper length predict body mass (m_1), compared to the assumption that only flipper length predicts body mass (m_0).

Combining relative & absolute evidence




```
mcmc_areas(m1, pars = c("b_bill_len",
"b_flipper_len", "sigma"))
```

The $BF_{10} \sim 20$ was observed although the credible interval for bill length included the Null, suggesting the impact of bill length was small in absolute terms.



Source: www.bayesianspectacles.org